

Додаток 2. Результати тестування

Basic

examples/basic.signal

```
1 PROGRAM HELLO;  
2 CONST  
3   X = '42';  
4 BEGIN  
5 END.
```

Output

Code	Row	Col	Type	Value
256	1	1	Keyword	PROGRAM
1000	1	9	Identifier	HELLO
59	1	14	Delimiter	;
257	2	1	Keyword	CONST
1001	3	3	Identifier	X
61	3	5	Delimiter	=
39	3	7	Delimiter	'
500	3	8	Literal	42
39	3	10	Delimiter	'
59	3	11	Delimiter	;
258	4	1	Keyword	BEGIN
259	5	1	Keyword	END
46	5	4	Delimiter	.

Comments

examples/comments.signal

```
1 (* A lot of comments *)
2 PROGRAM COMMENTS;
3 CONST
4   (* Single-line comment *)
5   (*
6     Multi-line comment
7   *)
8   (**) (* *)
9   (* Multiple *) (* comments *) (* on *) (* the *) (* same *) (* line *)
10  (****) ((*()*())*) (*;.:*) (* Comments with nested * and () *)
11  (* Invalid symbols are allowed! %#^@& *)
12 BEGIN (* on the same like as identifier *)
13 END (* between tokens *) .
```

Output

Code	Row	Col	Type	Value
256	2	1	Keyword	PROGRAM
1000	2	9	Identifier	COMMENTS
59	2	17	Delimiter	;
257	3	1	Keyword	CONST
258	12	1	Keyword	BEGIN
259	13	1	Keyword	END
46	13	26	Delimiter	.

Constants

examples/constants.signal

```
1 (* All ways to declare a constant within a grammar *)
2 PROGRAM CONSTANTS;
3 CONST
4   (* Single-line comment *)
5   X = '';
6   A = '1';
7   B = '2,3';
8   C = ',4';
9   D = '5$EXP(6)';
10  E = '$EXP(7)';
11 BEGIN
12   (*
13    Multi-line comment
14   *)
15 END.
```

Output

Code	Row	Col	Type	Value
256	2	1	Keyword	PROGRAM
1000	2	9	Identifier	CONSTANTS
59	2	18	Delimiter	;
257	3	1	Keyword	CONST
1001	5	3	Identifier	X
61	5	5	Delimiter	=
39	5	7	Delimiter	'
39	5	8	Delimiter	'
59	5	9	Delimiter	;
1002	6	3	Identifier	A
61	6	5	Delimiter	=
39	6	7	Delimiter	'
500	6	8	Literal	1
39	6	9	Delimiter	'
59	6	10	Delimiter	;
1003	7	3	Identifier	B
61	7	5	Delimiter	=
39	7	7	Delimiter	'
501	7	8	Literal	2
44	7	9	Delimiter	,
502	7	10	Literal	3
39	7	11	Delimiter	'
59	7	12	Delimiter	;
1004	8	3	Identifier	C
61	8	5	Delimiter	=
39	8	7	Delimiter	'
44	8	8	Delimiter	,
503	8	9	Literal	4
39	8	10	Delimiter	'
59	8	11	Delimiter	;
1005	9	3	Identifier	D
61	9	5	Delimiter	=
39	9	7	Delimiter	'
504	9	8	Literal	5
36	9	9	Delimiter	\$
260	9	10	Keyword	EXP
40	9	13	Delimiter	(
505	9	14	Literal	6
41	9	15	Delimiter	)
39	9	16	Delimiter	'
59	9	17	Delimiter	;

1006	10	3	Identifier	E
61	10	5	Delimiter	=
39	10	7	Delimiter	'
36	10	8	Delimiter	\$
260	10	9	Keyword	EXP
40	10	12	Delimiter	(
506	10	13	Literal	7
41	10	14	Delimiter	)
39	10	15	Delimiter	'
59	10	16	Delimiter	;
258	11	1	Keyword	BEGIN
259	15	1	Keyword	END
46	15	4	Delimiter	.

Edge Cases

examples/edge-cases.signal

```
1 (* Edge cases within the grammar *)
2 PROGRAM EDGES;
3 CONST
4   AB12CD = '999,000';
5   Z = '100$EXP(200)';
6   LONGNAME123 = '0';
7   A = '0,0';
8 BEGIN
9 END.
```

Output

Code	Row	Col	Type	Value
256	2	1	Keyword	PROGRAM
1000	2	9	Identifier	EDGES
59	2	14	Delimiter	;
257	3	1	Keyword	CONST
1001	4	3	Identifier	AB12CD
61	4	10	Delimiter	=
39	4	12	Delimiter	'
500	4	13	Literal	999
44	4	16	Delimiter	,
501	4	17	Literal	000
39	4	20	Delimiter	'
59	4	21	Delimiter	;
1002	5	3	Identifier	Z
61	5	5	Delimiter	=
39	5	7	Delimiter	'
502	5	8	Literal	100
36	5	11	Delimiter	\$
260	5	12	Keyword	EXP
40	5	15	Delimiter	(
503	5	16	Literal	200
41	5	19	Delimiter	)
39	5	20	Delimiter	'
59	5	21	Delimiter	;
1003	6	3	Identifier	LONGNAME123
61	6	15	Delimiter	=
39	6	17	Delimiter	'
504	6	18	Literal	0
39	6	19	Delimiter	'
59	6	20	Delimiter	;
1004	7	3	Identifier	A
61	7	5	Delimiter	=
39	7	7	Delimiter	'
504	7	8	Literal	0
44	7	9	Delimiter	,
504	7	10	Literal	0
39	7	11	Delimiter	'
59	7	12	Delimiter	;
258	8	1	Keyword	BEGIN
259	9	1	Keyword	END
46	9	4	Delimiter	.

Empty

examples/empty.signal

```
1 (* Empty program *)
2 PROGRAM EMPTY;
3 BEGIN
4 END.
```

Output

Code	Row	Col	Type	Value
256	2	1	Keyword	PROGRAM
1000	2	9	Identifier	EMPTY
59	2	14	Delimiter	;
258	3	1	Keyword	BEGIN
259	4	1	Keyword	END
46	4	4	Delimiter	.

Error Comment Not Closed

examples/error-comment-not-closed.signal

```
1 (* Check comment not closed error *)
2 PROGRAM NOCLOSE;
3 BEGIN
4 END.
5 (* No closing bracket!
```

Output

Code	Row	Col	Type	Value
256	2	1	Keyword	PROGRAM
1000	2	9	Identifier	NOCLOSE
59	2	16	Delimiter	;
258	3	1	Keyword	BEGIN
259	4	1	Keyword	END
46	4	4	Delimiter	.
Tokenizer Error [5:1]: Comment not closed				

Error Invalid Character

examples/error-invalid-character.signal

```
1 (* Encountered invalid character(s) *)
2 PROGRAM INVCHARS;
3 CONST
4   A = '1,%'; (* 1 error: % is invalid *)
5   b#& = '30$EXP(40)'; (* 3 errors: b, #, & are invalid *)
6 BEGIN (* 1 error: grammar is case-sensitive! *)
7 END.
```

Output

	Code	Row	Col	Type	Value
	256	2	1	Keyword	PROGRAM
	1000	2	9	Identifier	INVCHARS
	59	2	17	Delimiter	;
	257	3	1	Keyword	CONST
	1001	4	3	Identifier	A
	61	4	5	Delimiter	=
	39	4	7	Delimiter	'
	500	4	8	Literal	1
	44	4	9	Delimiter	,
	39	4	11	Delimiter	'
	59	4	12	Delimiter	;
	61	5	7	Delimiter	=
	39	5	9	Delimiter	'
	501	5	10	Literal	30
	36	5	12	Delimiter	\$
	260	5	13	Keyword	EXP
	40	5	16	Delimiter	(
	502	5	17	Literal	40
	41	5	19	Delimiter	)
	39	5	20	Delimiter	'
	59	5	21	Delimiter	;
	1002	6	1	Identifier	BEGI
	259	7	1	Keyword	END
	46	7	4	Delimiter	.
Tokenizer	Error [4:10]: Invalid character '%'				
Tokenizer	Error [5:3]: Invalid character 'b'				
Tokenizer	Error [5:4]: Invalid character '#'				
Tokenizer	Error [5:5]: Invalid character '&'				
Tokenizer	Error [6:5]: Invalid character 'n'				

Error Mixed

examples/error-mixed.signal

```
1 (* Multiple error types in one file *)
2 PROGRAM MIXED;
3 CONST
4   A = '1';
5   bad = '2'; (* lowercase: 3 errors *)
6   C = '3~4'; (* invalid ~: 1 error *)
7 BEGIN
8 END.
9 (* unclosed comment
```

Output

Code	Row	Col	Type	Value
256	2	1	Keyword	PROGRAM
1000	2	9	Identifier	MIXED
59	2	14	Delimiter	;
257	3	1	Keyword	CONST
1001	4	3	Identifier	A
61	4	5	Delimiter	=
39	4	7	Delimiter	'
500	4	8	Literal	1
39	4	9	Delimiter	'
59	4	10	Delimiter	;
61	5	7	Delimiter	=
39	5	9	Delimiter	'
501	5	10	Literal	2
39	5	11	Delimiter	'
59	5	12	Delimiter	;
1002	6	3	Identifier	C
61	6	5	Delimiter	=
39	6	7	Delimiter	'
502	6	8	Literal	3
503	6	10	Literal	4
39	6	11	Delimiter	'
59	6	12	Delimiter	;
258	7	1	Keyword	BEGIN
259	8	1	Keyword	END
46	8	4	Delimiter	.
Tokenizer Error [5:3]: Invalid character 'b'				
Tokenizer Error [5:4]: Invalid character 'a'				
Tokenizer Error [5:5]: Invalid character 'd'				
Tokenizer Error [6:9]: Invalid character '~'				
Tokenizer Error [9:1]: Comment not closed				